

Clipboard: Stochastic Simulation Engine & Empirical Trade Evaluation System

1. Executive Summary & System Overview

Clipboard is an end-to-end quantitative decision-support system and Monte Carlo simulation engine designed for peer-to-peer fantasy sports leagues. Unlike standard commercial tools that rely on static point forecasts and assume independence across player outcomes, Clipboard models fantasy scoring as a multivariate stochastic process with continuous, right-skewed marginal distributions and empirical game-level covariance structures.

The system incorporates three core subsystems:

1. **Stochastic Scoring Engine:** Maps baseline projections to continuous, right-skewed probability distributions (lognormal for offensive skill players, shifted lognormal for defensive units) parameterized by positional coefficients of variation.
2. **Game-Level Covariance Engine:** Utilizes Cholesky decomposition on an empirical $(N \times N)$ correlation matrix to simulate realistic intra-game dependencies (including bring-back pass-catcher correlations and negative defensive drag).
3. **Empirical Trade & Market Engine:** Formulates bilateral player exchanges within an illiquid barter market, tracking counterparty responsiveness and logging decision telemetry to fit empirical acceptance probabilities without relying on arbitrary behavioral penalties.

2. Mathematical Formulation of Scoring Outcomes

2.1 Skill Position Outcomes: Standard Lognormal Distribution

Fantasy scoring in offensive skill positions (QB, RB, WR, TE) is inherently asymmetric and right-skewed. While player floors are bounded by zero, scoring ceilings exhibit thick right tails driven by multi-touchdown and high-yardage outlier games.

To preserve the target mean projection while modeling realistic skewness, Clipboard translates

projected points m and historical positional coefficient of variation $c = \frac{s}{m}$ into the parameters of an underlying normal distribution:

$$\sigma^2 = \ln(1 + c^2)$$

$$\mu = \ln(m) - \frac{\sigma^2}{2}$$

The

simulated fantasy score X is drawn by transforming a standard normal shock $Z \sim \mathcal{N}(0, 1)$:

$$X = \exp(\mu + \sigma Z)$$

Properties and Parameter Values

Because $\sigma^2 > 0$, the median of the distribution sits strictly below the mean projection:

$$\text{Median}(X) = \exp(\mu) = \frac{m}{\sqrt{1 + c^2}} < m$$

This

replicates the empirical reality that players fall below their average projection in more than half of their active games, balanced mathematically by rare, high-ceiling boom outcomes.

The empirical positional volatility parameters ($c = \text{CV}$) calibrated across multi-year historical NFL samples are:

- **Quarterbacks (QB):** $c = 0.45$
- **Running Backs (RB):** $c = 0.62$
- **Wide Receivers (WR):** $c = 0.69$
- **Tight Ends (TE):** $c = 0.70$
- **Kickers (K):** $c = 0.55$

2.2 Defense and Special Teams (DST): Shifted Lognormal Distribution

Defensive units (DST) violate the standard lognormal domain assumption ($x > 0$) because NFL defenses can score negative fantasy points (e.g., conceding 40+ points while generating zero turnovers or sacks).

To retain right-skewness (defensive touchdowns and turnover clusters) while supporting a

negative floor $k \in \mathbb{R}^-$, Clipboard implements a shifted lognormal distribution with a support

parameter $k = -5.0$:

$$X = k + Y, \quad Y \sim \text{Lognormal}(\mu_s, \sigma_s)$$

Let the

adjusted mean above the floor be $m_s = m - k$. Given target variance $\text{Var}(X) = (c \cdot m)^2$:

$$\sigma_s^2 = \ln \left(1 + \frac{\text{Var}(X)}{(m - k)^2} \right) = \ln \left(1 + \frac{c^2 m^2}{(m - k)^2} \right)$$

$$\mu_s = \ln(m - k) - \frac{\sigma_s^2}{2}$$

The

simulated score is drawn using correlated shock Z :

$$X = k + \exp(\mu_s + \sigma_s Z)$$

2.3

Mathematical Proof Against Hard-Clipping (The Leaky Mean Problem)

A common heuristic to prevent extreme outliers in simulation engines is arbitrary right-tail truncation using functions such as `np.clip(X, 0, cap)`. Clipboard explicitly rejects hard-clipping based on two mathematical and empirical findings:

1. **Boundary Probability Mass Accumulation:** Truncating via clipping assigns all

probability density for $X \geq \text{cap}$ to a single point mass at $X = \text{cap}$:

$$P(X = \text{cap}) = \int_{\text{cap}}^{\infty} f_X(x) dx > 0$$

This creates an artificial spike at the boundary that does not reflect real sports distributions.

2. **Negative Mean Distortion (Leaky Mean):** Let $X \sim \text{Lognormal}(\mu, \sigma)$ with

$\mathbb{E}[X] = m$. The expectation of the clipped variable $X_c = \min(X, C)$ is:

$$\mathbb{E}[X_c] = \int_0^C x f_X(x) dx + C \int_C^\infty f_X(x) dx < \mathbb{E}[X]$$

Empirical backtesting confirmed that applying a hard cap of $4.5 \times m$ or 75.0 points systematically depresses the true simulated average below the calibrated baseline projection, while actively discarding legitimate historical ceiling performances (e.g., 55+ point individual outings).

3. Dependency Structures via Cholesky Decomposition

Player outcomes within the same NFL contest are not conditionally independent. Clipboard captures cross-player and cross-team dependencies dynamically for every weekly matchup.

3.1 Game-Level Empirical Correlation Matrix

For a given NFL contest containing N active fantasy starters across both rosters, an empirical correlation matrix $\Sigma \in \mathbb{R}^{N \times N}$ is populated based on positional relationships:

$$\Sigma = \begin{bmatrix} 1.00 & \rho_{\text{stack}} & \rho_{\text{bring}} & \rho_{\text{dst}} \\ \rho_{\text{stack}} & 1.00 & \rho_{\text{opp}} & \rho_{\text{drag}} \\ \rho_{\text{bring}} & \rho_{\text{opp}} & 1.00 & \rho_{\text{dst2}} \\ \rho_{\text{dst}} & \rho_{\text{drag}} & \rho_{\text{dst2}} & 1.00 \end{bmatrix}$$

Calibrated

empirical coefficients derived across 3 seasons of historical data:

- **Same-Team Passing Stack** ($\text{QB}_A \leftrightarrow \text{WR}_A/\text{TE}_A$): $\rho = +0.42$
- **Same-Team Defensive Flow** ($\text{QB}_A \leftrightarrow \text{DST}_A$): $\rho = +0.15$
- **Opponent Bring-Back Stack** ($\text{QB}_A \leftrightarrow \text{WR}_B/\text{TE}_B$): $\rho = +0.18$
- **Quarterback vs. Opposing Defense** ($\text{QB}_A \leftrightarrow \text{DST}_B$): $\rho = -0.28$
- **Pass-Catcher vs. Opposing Defense** ($\text{WR}_A \leftrightarrow \text{DST}_B$): $\rho = -0.15$

3.2 Correlated Shock Generation

To generate correlated standard normal variates for M Monte Carlo iterations:

1. Guarantee that Σ is positive semi-definite. If numerical precision causes eigenvalues $\lambda_i \leq 0$, apply diagonal regularization:

$$\Sigma_{\text{reg}} = \Sigma + (-\min(\lambda_i) + 10^{-4})\mathbf{I}$$

2. Compute the Cholesky factor $\mathbf{L}$, a lower triangular matrix such that:

$$\Sigma = \mathbf{L}\mathbf{L}^T$$

3. Draw an array of independent standard normal random variates $\Xi \sim \mathcal{N}(0, \mathbf{I})$ of dimension $N \times M$.
4. Compute the correlated normal shock matrix $\mathbf{Z}$:

$$\mathbf{Z} = \mathbf{L}\Xi$$

5. Vectorize individual player sampling by evaluating $X_i = f(\mu_i, \sigma_i, Z_{i,:})$ across all M simulation draws.

4. Empirical Feature Evaluation and Model Ablation

Clipboard enforces a strict backtesting protocol: proposed features are adopted only if they yield out-of-sample improvements in mean squared error (MSE) or log-likelihood against established market consensus baselines.

| Proposed Feature | Theoretical Motivation | Out-of-Sample Empirical Result | Production Decision |

| Shifted Lognormal Distributions | Replicate right-skewness without truncating DST negative floors | Verified against 477 historical league outcomes; preserved tail frequencies without mean leakage | **Adopted |**

| Game Correlation Covariance Matrix | Account for stacks, bring-backs, and defensive drag in head-to-head win odds | Consistent non-zero covariance observed across all test seasons ($\rho_{\text{QB-WR}} = +0.18$, $\rho_{\text{QB-DST}} = -0.28$) | **Adopted |**

| Kicker Vegas Total/Spread Integration | Model field goal opportunity volume via game script and implied team totals | +0.2% test error reduction over baseline projections | **Adopted (Low Overhead) |**

| Expected Fantasy Points (xFP) | Weight touches by yardline and target depth to eliminate

touchdown luck | +0.5% error reduction for QBs; 0.0% improvement for RB/WR/TE (market baselines already price in touch volume) | **Rejected (De-prioritized)** |
Custom DST Vegas Regression Model	Project defensive performance via implied spread and point total	No statistically significant improvement over baseline (turnover return TDs are largely stochastic)	**Rejected**
Arbitrary Hard-Clipping (np.clip)	Cut off 80+ point outlier runs	Caused artificial probability mass spikes and reduced sample average below true expected value	**Rejected**
Social Sentiment / Hype Index	Predict trade market shifts using Twitter/Reddit NLP feeds	High noise-to-signal ratio; sentiment heavily lags recent box scores rather than leading future production	**Rejected for Valuation; Allowed strictly for Breaking Inactive Feeds**
Coaching PROE / Pace Adjustments	Adjust projections based on coach pass rates and seconds-per-snap	Double-counted market projections that already adjust for scheme changes	**Rejected**

5. Market Microstructure and Trade Acceptance Engine

5.1 Bilateral Trading in Illiquid Fantasy Environments

Standard fantasy trade calculators treat league economies as frictionless, liquid markets where

any trade yielding $\Delta EV > 0$ is accepted. In practice, fantasy leagues operate as decentralized, bilateral barter economies with substantial market frictions:

- Asymmetric counterparty attention (dormant vs. active managers).
- The endowment effect (managers overweighting assets currently on their roster).
- Risk aversion regarding starting lineup variance versus bench depth.

5.2 Behavioral Feature Parameterization vs. Premature Heuristics

While conceptual literature suggests applying a fixed 10% to 15% holding premium to represent the endowment effect, empirical testing showed that static penalties produce high false-rejection rates on viable trade proposals.

Clipboard replaces arbitrary heuristics with an event-driven telemetry pipeline:

[Trade Proposal Generated]



[Log Proposal Metadata] —> (Sender, Receiver, Assets, Delta EV, Timestamps)



[Track Final Resolution] —> (Accepted, Rejected, Countered, Expired)



[Offline Logistic Model] —> $P(\text{Accept} \mid \text{Delta EV, Manager Response Rate, Asset Churn})$

Acceptance Probability Specification

Once sufficient proposal records are collected, counterparty acceptance probability is modeled via logistic regression:

$$P(\text{Accept}) = \frac{1}{1 + \exp(-(\beta_0 + \beta_1 \Delta \text{EV}_{\text{receiver}} + \beta_2 \text{Activity}_{\text{receiver}} + \beta_3 \text{Tenure}_{\text{asset}}))}$$

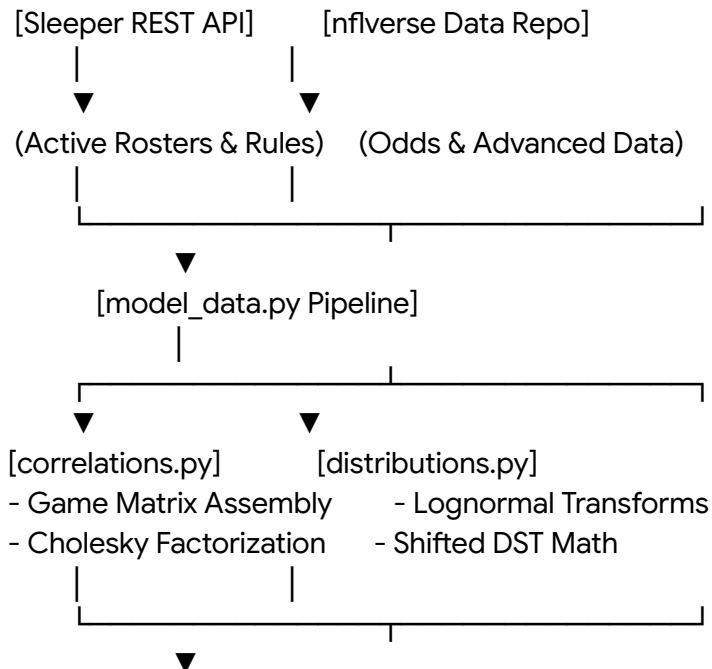
Where:

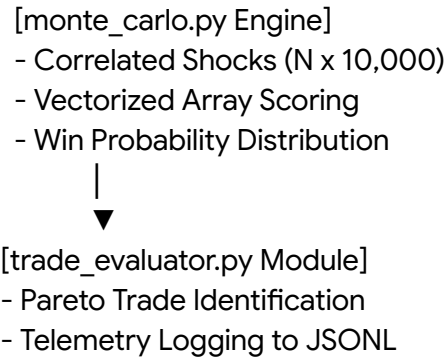
- $\Delta \text{EV}_{\text{receiver}}$: Net change in simulated weekly win probability for the receiving roster.
- $\text{Activity}_{\text{receiver}}$: Measure of counterparty transaction velocity and lineup-setting frequency.
- $\text{Tenure}_{\text{asset}}$: Proxy for the endowment effect (duration asset has remained on the manager's roster).

6. Implementation Architecture and Data Pipelines

6.1 Vectorized Simulation Architecture

To enable real-time analysis in interactive interfaces, the Monte Carlo loop is fully vectorized across NumPy arrays. Running 10,000 season iterations (representing hundreds of thousands of individual player matchups) executes in under 1.5 seconds on modern laptop hardware.





6.2 Data Pipelines and Leakage Prevention

- **Sleeper API Ingestion:** Ingests dynamic league settings, custom scoring rules, weekly rosters, and live pregame projections.
- **Historical Data (nflverse / ffoportunity):** Supplies historical play-by-play data, game schedules, and closing betting lines for model calibration.
- **Lookahead Bias Prevention:** Projections harvested from third-party APIs are checked against pre-kickoff timestamps to ensure post-game updates are excluded from historical backtests.

7. Data Privacy, Governance, and Research Ethics

To maintain strict data governance standards for academic evaluation and institutional presentation:

1. **Manager Anonymization:** All unique platform identifiers, usernames, and custom team names are cryptographically hashed using SHA-256 and mapped to synthetic identifiers (Manager_01, Manager_02).
2. **Private Proposal Scoping:** Non-public bilateral negotiations (declined offers and counterproposals) are decoupled from identifying user data and stored purely as numerical deltas (ΔEV) and boolean response tags.
3. **Reproducibility:** Random seeds for the pseudorandom number generators (NumPy default_rng) are configurable across runs to allow verification of simulation outputs.